

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Engineered a pluggable distributed executor routing workloads across Docker and AWS Lambda with an S3 artifact store, eliminating redundant recomputation at scale.
- Own production services end-to-end, architecture through deployment, monitoring, and on-call, handling millions of daily transactions at 99%+ uptime.
- Debugged and resolved defects across a large, fast-moving production codebase, estimating work complexity and designing integration strategies for architectural changes.
- Participated in design and code reviews setting rigorous code quality standards across the platform.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Extended and optimized inference serving internals with kernel-level development (CUDA/Triton), distributed KV cache, and speculative decoding to improve serving performance.
- Cut p99 serving latency 420ms to 140ms across 50K+ daily requests through systems-level performance analysis and an async serving redesign.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated 25+ production C++ and SQL-backed distributed systems for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Cut nightly financial data aggregation blocking downstream settlement from 8 minutes to 90 seconds by redesigning SQL with indexed views and batch partitioning.
- Standardized failure-handling contracts across 25+ financial services, reducing production escalations 60% in one quarter via consistent retry, timeout, and propagation semantics.

PROJECTS

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Built a Raft-consensus distributed storage engine from scratch in C++: concurrent writes, linearizable reads, and log compaction, a hands-on study of database internals and consensus tradeoffs.
- Built a companion VPC control plane simulator modeling subnet routing and resource placement over the same distributed substrate.

LangFetch

tinyurl.com/LangFetch

- Agent decomposing multi-table questions into dependency-ordered SQL subqueries with intermediate result passing and query planning across relational data sources.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Full-stack platform with FastAPI backend, React/TypeScript dashboard, and PostgreSQL, exposing APIs for rubric-based scoring and latency tracking at scale.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: C++, Java, Python, TypeScript, SQL

Systems & Storage: Distributed Storage Systems, Consensus Protocols, Database Internals, Query Planning, Concurrency

Performance: CUDA/Triton Kernel Development, Low-Level Performance Analysis, Distributed KV Cache

Infrastructure: AWS (EC2, S3, RDS, SQS), GCP (Cloud Run), Docker, Kubernetes, CI/CD

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer